

Chaitanya.D

Software Architect / Swedish Educated / Building systems across telecom, networking & cloud

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Skills

C++11/14/17/20/23 | Agentic Engineering | Golang | Python | Docker | Kubernetes | Agentic AI | REST API | LangChain | LangGraph | LangSmith | PostgreSQL | TimescaleDB | SQLite | TuringDB | Databases | Graph databases | Microservices Architecture | Git | jj-vcs | VOIP | SIP | Network Packet Analysis | WireShark | Data Analytics | DSL Parsers (ANTLR) | System Design | OOP Design Patterns | Functional Programming Patterns | CI/CD Pipelines | Jenkins | MATLAB | Mathematica | Bluetooth Stack | Technical Leadership | AWS | C | Elixir | Distributed and Scalable Systems | Cross-functional team collaboration | GNU / Linux | Debugging / troubleshooting

Work Experience

Spirent (VIAVI), Staff Software Engineer (*Architect*) Bangalore, India

2019 - Present

- Contributed as an architect to a cross-functional and geographically distributed team that implemented performance enhancements to scale the existing application, driving \$1M+ in additional revenue.
- Enhanced the network test result analysis product by implementing massive-scale (scaling from 64K streams to 10M flows) autonomous flow tracking feature along with real-time event correlation.
- Feature evolution of a high-throughput historical time-series data engine along with production support.
- Engineered a cross-platform log aggregation tool that ingests unstructured logs from distributed services across Windows, Linux, and Chassis nodes, transforming and streaming them as structured data to a centralized location.
- Extracted business domain logic from monolithic system (C#, C++) and architected corresponding micro-services applications (Golang, Docker) for improved scalability and maintainability.
- Pioneered Golang adoption in the India dev center as the first engineer to contribute to a US-led modernization initiative, paving the way for the broader team's successful transition to the modernized platform.

- Worked on legacy full stack composing of C# UI, C++ middle-ware & Python hardware layer of the existing application to deliver a feature to obsolete legacy hardware within a few months of joining the organization.
- Mentored junior engineer to get up to speed with the existing application architecture and code structure. Helped them to get going on bug fixing and feature development.

Lead Software Engineer, Infinera (Nokia) Bangalore, India

2017 - 2018

- Implemented L2 switching features leveraging Broadcom SDK for data link layer network operations.
- Extended functionality of the optical chassis simulator, adding control plane capabilities for optical networking.
- Took initiative, beyond primary duties, to contribute to a NodeJS diagnostic tool for chassis troubleshooting.

System Analyst, Actiance (Smarsh) Bangalore, India

2017 - 2017

- Resolved critical instability issues and performance bottlenecks in the STUN module within six months of joining the team & contributed significantly to developing a recording solution for Skype for Business.
- Worked closely with the performance team to profile, optimize, and ultimately achieve a remarkable 30% reduction in memory usage and a significant 40% improvement in execution speed across the application.

Senior Software Engineer, UMS (Everbridge) Bangalore, India & Stavanger, Norway

2013 - 2017

- Architected and delivered back-end components for the IVR application and led end-to-end development of a call processing library for Dialogic HMP (E1/T1) telecom cards, applying advanced C++ design patterns (policy based design), C++11 features, STL, and Boost libraries while resolving critical production issues.
- Partnered with architects, remote teams, and sales across all phases — from product specification reviews, Developing POCs and project estimation along with managing production MS-SQL database servers.

Research Engineer, Fraunhofer AISEC Munich, Germany

2011 - 2011

- Completed Master's thesis (Topic: Networking for smart meters) funded by German BSI.

Software Engineer, Nokia Siemens Networks (Nokia) Helsinki, Finland

2010 - 2010

- Configuration of complex network test environments & unit testing the IPSec module in C++.

Software Engineer, MindTree (LTM) Bangalore, India

2007 - 2009

- Designed, implemented, and ported Bluetooth profiles and protocols across multiple embedded systems with proprietary RTOS (Car Infotainment Kits), including audio codec integration and security auditing.

Education

Master of Science in Communication Systems

Kungliga Tekniska Högskolan Stockholm, Sweden

Bachelor of Engineering in Telecommunication Engineering

P.E.S Institute of Technology, Visveswaraiah Technological University Bangalore, India

Interests

Running | Reading novels by Dan Brown | Traveling.